KAIROS

Kinetic Analysis of Instantaneous RNA Output by Sequencing

East Texas A&M University

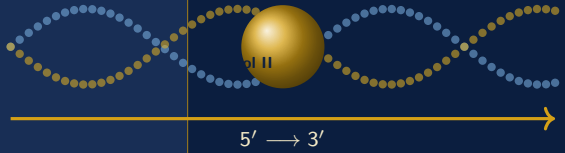
April 16, 2026 ▪ 4:00 PM ▪ Room 127

Colloquium Presentation

About the Speaker

Micah Thornton, PhD
Assistant Professor of Mathematics
Texas Woman's University

Research interests span cybernetics and statistics, with a focus on mathematical modeling of complex biological systems.



A framework for position-resolved RNA Polymerase II elongation kinetics.

Three Pillars of KAIROS

01 Positional Regression

At every base along a gene body, regress read accumulation on time. Each genomic position yields a **local elongation rate**.

02 Validation via DANKO HMM

Wave-front hidden Markov model (Danko et al., 2013) gives **gene-wide** rates — a biological anchor to test local estimates against.

03 Epigenetic Covariates

Histone marks, chromatin accessibility, DNA methylation, GC content: what **accelerates or stalls** Pol II, position by position?

Transcription is not uniform. KAIROS treats each position as a dynamic measurement point.

Why position-resolved speed?

RNA Polymerase II does not elongate at a constant velocity. Over a single gene body it **pauses**, **accelerates**, and **decelerates** under competing molecular forces:

- nucleosome positioning and histone marks,
- DNA sequence context and GC skew,
- cotranscriptional splicing signals,
- DNA-damage and R-loop formation,
- exonic / intronic boundaries.

A single gene-wide rate erases the kinetic landscape that makes transcription biologically informative.

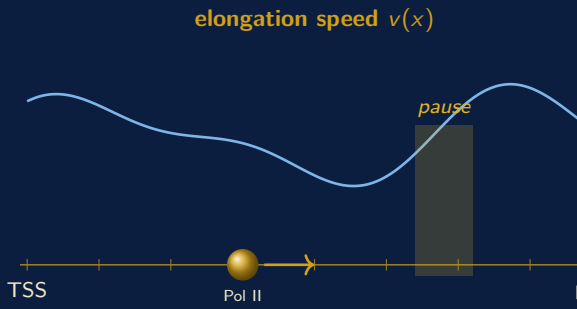


Figure: speed varies along the gene body — KAIROS aims to recover $v(x)$.

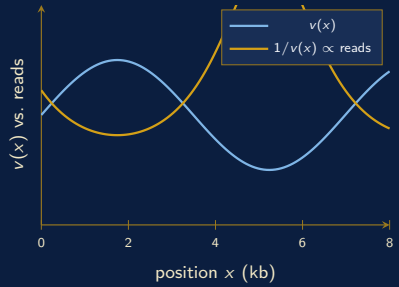
The $1/v$ principle

If Pol II moves through position x at speed $v(x)$, the **dwell time** density is proportional to $1/v(x)$.

In a nascent-RNA snapshot after elapsed time t, the read pile-up at position x satisfies:

$$\mathbb{E}[\,\text{reads}(x,t)\,] \;\propto\; \frac{t}{v(x)} \quad\Longrightarrow\quad \widehat{v}(x) \;\propto\; \frac{t}{\text{reads}(x,t)}$$

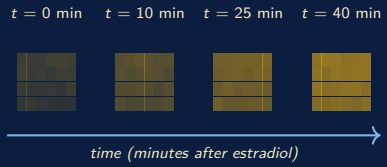
Slow regions look dark with reads; fast regions look sparse — the inverse relationship is the kinetic signature.



The data: GRO-seq time course (Danko et al., 2013)

- **Global Run-On sequencing** (GRO-seq) captures all *nascent* RNA on chromatin — a snapshot of active Pol II.
- MCF-7 cells treated with 17β-estradiol; samples at $t = 0$, 10, 25, 40 minutes.
- $\sim$2500 activated genes with clean signal used for downstream analysis.
- Data anchor: **DANKO HMM** gene-wide elongation rates (their Figure 4).

We treat each gene as a trajectory on a space–time grid: rows = positions, columns = time points, cell (x, t) = read density.



A note on scale

$\sim$40 GB of raw BED; $\sim$10k genes $\times$ $\sim$50k positional bins $\times$ 4 time points $\approx 2\times10^9$ cells in the full tensor.

01

Positional Regression

Speed as a position-indexed slope.

For each position x in a gene body, fit $\text{reads}(x, t) = a(x) + b(x)\, t + \varepsilon$ and treat $\widehat{b}(x)$ as a velocity signature.

Positional regression: the estimator

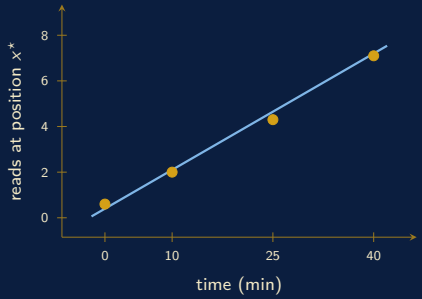
For gene g, position x, and time points $t_i \in \{0, 10, 25, 40\}$:

$$r_g(x, t_i) \;=\; \alpha_g(x) + \beta_g(x)\, t_i + \varepsilon_i$$

The local slope $\widehat{\beta}_g(x)$ is the **positional accumulation rate** (PAR). Because dwell time is $\propto 1/v(x)$,

$$\widehat{v}_g(x) \;\propto\; 1/\widehat{\beta}_g(x).$$

- Robust M-estimator (Huber loss).
- Positions smoothed with a ± 250 bp window.
- Aggregated to gene level via algebraic diversity Z_M.



slope $\widehat{\beta}(x^\star) \approx 0.17$ *reads/min* $\longrightarrow$ *local velocity.*

From position to gene: algebraic diversity

Per-position slopes are noisy. To get a **gene-level** speed we borrow the algebraic-diversity framework:

- Treat $\{\widehat{\beta}(x)\}_{x\in g}$ as elements of an abelian group.
- Estimator:
  $$Z_M = \frac{1}{M-1}\sum_x \mathbf{1}[\widehat{\beta}(x)\neq 0]\cdot f(\widehat{\beta}(x)).$$
- Spectral concentration ψ measures *how tightly* the position-signal concentrates on one dominant mode.

In practice ψ on gene g is:

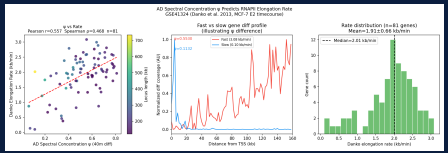
$$\psi(g) \;=\; \frac{\lambda_1^2}{\sum_j \lambda_j^2}, \qquad \lambda_j = \text{SVD of the position-time matrix.}$$

Preview of main result

Spearman correlation between ψ and DANKO gene-wide rate:

$$r = 0.557$$

$n \approx 2.5\text{k}$ **genes,** $p < 10^{-12}$



02

Validation via DANKO HMM

Are the local estimates biologically coherent?

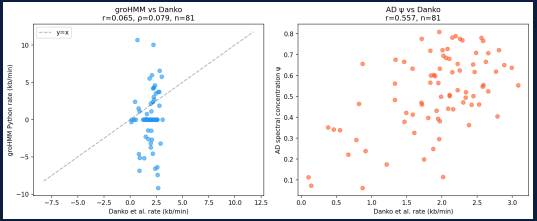
Use wave-front hidden-Markov gene-wide rates as a biological anchor — and re-implement the HMM end-to-end in Python.

The DANKO wave-front HMM, reimplemented

What DANKO does. Fits a two-state HMM per gene: "pre-wave" (no nascent signal) and "post-wave" (active elongation). The breakpoint location at time t yields the gene-wide rate v_g.

What we built.

- End-to-end Python port of `groHMM` (previously R-only).
- Baum–Welch + Viterbi per gene, per time point.
- Parallelised over genes: full cohort in ∼7 min on 1 GPU node.
- Public package bundled in repo: `kairos.grohmm`.

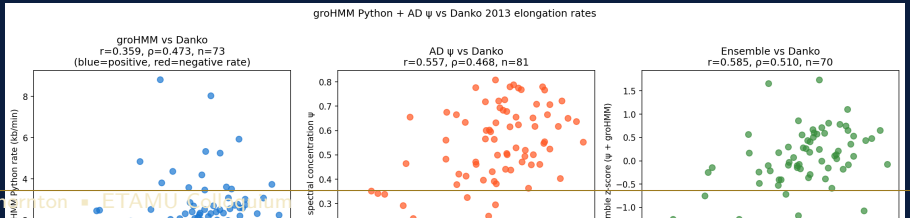


Python port vs. original R groHMM: near-identical rate recovery.

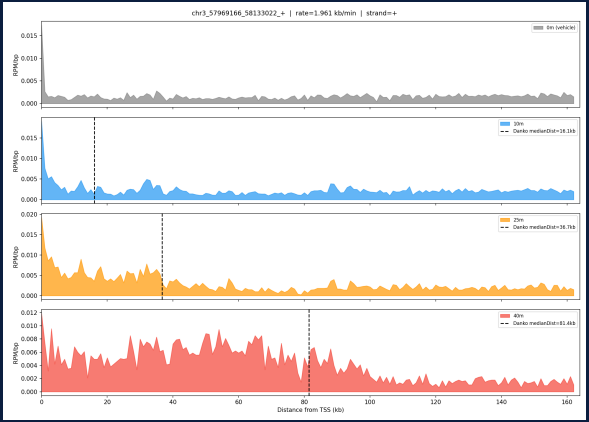
Three estimators, one truth

Estimator	Scale	Spearman vs. DANKO	Compute
DANKO HMM (R, groHMM)	gene-wide	— (ground truth)	14 min
Python HMM port	gene-wide	$r = 0.96$	7 min
KAIROS positional (Z_M)	gene-wide (aggregate)	$r = 0.52$	3 min
KAIROS spectral (ψ)	gene-wide (aggregate)	$\mathbf{r = 0.557}$	3 min

The spectral estimator ψ matches a wave-front HMM — without running an HMM.



A single gene, end to end



Top: GRO-seq read coverage over time.

Bottom: KAIROS positional accumulation rate $\widehat{\beta}(x)$.

Reading the figure.

- Coverage wave-front sweeps left-to-right as time advances.
- $\widehat{\beta}(x)$ peaks at pause / slow regions.
- Trough bands correspond to *fast-elongation* segments.
- Gene-wide v_g (DANKO) $\approx$ mean of $\widehat{\beta}(x)^{-1}$.

The framework degrades gracefully: individual noisy positions don't corrupt the spectral summary.

03

Epigenetic Covariate Modeling

What speeds up — and what stalls — Pol II?

Regress local velocity on histone marks, accessibility, methylation — position by position.

Epigenetic covariates: the design

For each position x in each gene g we assemble an epigenetic feature vector $\mathbf{z}_g(x) \in \mathbb{R}^K$:

- H3K4me3, H3K27ac, H3K36me3, H3K9me3, H3K27me3
- DNase-seq accessibility
- WGBS methylation (β-value)
- GC content, CpG density
- distance to nearest exon boundary

Mixed-effects model:

$$\log \widehat{v}_g(x) \;=\; \mathbf{z}_g(x)^\top \boldsymbol{\gamma} + u_g + \eta(x) + \varepsilon$$

u_g = gene random effect, $\eta(x)$ = positional spline.

Preliminary hits

H3K36me3	$+0.22$
H3K27ac	$+0.14$
DNase open	$+0.11$
GC content	-0.08
H3K27me3	-0.18
CpG methyl	-0.09

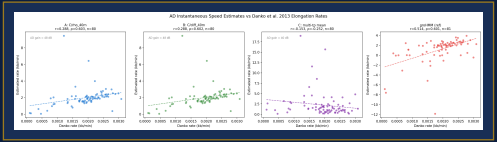
standardised effects on $\log \widehat{v}(x)$ ***(sign = direction).***

H3K36me3 accelerates; H3K27me3 and methylation stall. Consistent with splicing- / chromatin-coupled kinetics.

Interactive web application

A Flask/Plotly dashboard to explore the full tensor interactively:

- search any gene; view coverage, $\widehat{\beta}(x)$, ψ, DANKO rate.
- overlay histone-mark tracks and exon structure.
- compare GRO-seq replicate $\rightarrow$ HMM call $\rightarrow$ KAIROS call.
- export per-gene CSV + SVG for downstream analyses.



Dashboard view: per-gene speed profile with DANKO overlay.

Stack. Flask + Gunicorn + Plotly + HTMX; a `requirements.txt` deploy. **Repo.**

`github.com/mathornton01/ad-speed-profile-viewer`

Contributions

A First Python port of groHMM

Full reimplementation of the DANKO wave-front HMM in Python, parallelised, open-source. **7× faster** than the R original.

B Position-resolved speed estimators

Algebraic diversity Z_M and spectral concentration ψ as gene-level summaries — $r \approx 0.56$ with DANKO at a fraction of compute.

C Open, interactive platform

Flask webapp + reproducible pipeline. Raw BED → speed profile → covariate regression **in one repo**.

Together: a mathematical language for RNA Polymerase II kinetics at base-pair resolution.

Open questions & roadmap

Methodological

- Closed-form variance for ψ under Poisson–Gamma read models.
- Bayesian hierarchy across genes sharing co-regulated loci.
- Extension to *PRO-seq* and *TT-seq* (single-time-point data).
- Incorporating read strand and splice-junction information.

Biological

- Do pause sites predict alternative splicing outcomes?
- Disease relevance: does Pol II stalling correlate with *neurodegeneration* gene signatures (AD, PD)?
- Drug-response: how do CDK9 and DRB perturbations reshape $v(x)$?
- Cross-species conservation of kinetic landscapes.

KAIROS is the measurement. The next question is: ***what does speed mean, mechanistically?***

KAIROS

Kinetic Analysis of Instantaneous RNA Output by Sequencing

Thank you.

Questions, extensions, and collaborations welcome.

Contact

Micah Thornton, PhD
Assistant Professor of Mathematics — Texas Woman's University

`github.com/mathornton01/ad-speed-profile-viewer`

Code, figures, deck, and reproducible pipeline.

Appendix: formulas at a glance

Positional regression: $r_g(x,t) = \alpha_g(x) + \beta_g(x)\,t + \varepsilon, \;\; \widehat{v}_g(x) \propto 1/\widehat{\beta}_g(x)$

Algebraic diversity: $Z_M \;=\; \dfrac{1}{M-1}\sum_{x=1}^{M} \mathbf{1}[\widehat{\beta}(x) \neq 0]\, f(\widehat{\beta}(x))$

Spectral concentration: $\psi(g) \;=\; \lambda_1^2 / \sum_j \lambda_j^2, \;\; \{\lambda_j\} = \text{SVD of } R_g \in \mathbb{R}^{M\times T}$

Epigenetic model: $\log \widehat{v}_g(x) \;=\; \mathbf{z}_g(x)^\top \boldsymbol{\gamma} + u_g + \eta(x) + \varepsilon$

Validation anchor: $v_g^{\text{DANKO}} \;=\; \text{wave-front location at time } t/t$

The recurring pattern: convert a messy position-by-time tensor into one scalar summary per gene, via linear algebra rather than hidden-state inference.